

Review

Identifying energy-intensive key sectors in Iran: Evidence from decomposed input-output multipliers

Ali Faridzad ^{a,*}, Ali Asghar Banouei ^c, Jillian Banouei ^b, Zeinab Golestan ^a^a Department of Energy, Agriculture and Environment, Faculty of Economics, Allameh Tabataba'i University, Tehran, Iran^b Retired Faculty Member, ECO College of Insurance, Allameh Tabataba'i University, Tehran, Iran^c Department of Economic Development and Planning, Faculty of Economics, Allameh Tabataba'i University, Tehran, Iran

ARTICLE INFO

Article history:

Received 9 October 2018

Received in revised form

6 June 2019

Accepted 29 September 2019

Available online 30 September 2019

Handling Editor: Zheming Tong

JEL classification:

Q43

Q48

C67

Keywords:

Energy intensity

Own and pure multipliers

Input-output

Iran

ABSTRACT

The purpose of this study is to two-fold. One is to identify key economic sectors that contribute to fossil energy consumption of the Iranian economy and second is to suggest an energy management policy for cleaner production. The main question of this article is how to identify the sector which can reduce energy intensity and also is directly responsible for energy consumption. The energy intensity of each sector according to the Input-Output approach can be decomposed into own weighted component made during the activities of the sector and pure weighted component generated by the activities in other sectors. Based on Input-Output Table of Iran in 2011, the results show that six sectors including production and distribution of electricity, crude oil and natural gas extraction, transportation, manufacture of chemicals and chemical products, services and manufacture of other non-metallic mineral products are energy intensive key sectors of Iran which have the most value of own weighted backward and forward multipliers compared to the other economic sectors. Hence, appropriate policy implications are suggested for those six sectors. These policies should be targeted to the sectors with high own weighted effects like converting natural gas-based utilities into combined cycle ones in electricity sector. In the opposite case, since more than one sector is involved in fossil energy consumption, this type of direct policies cannot be recommended for those sectors with high pure weighted effects. Consequently, policy makers should employ cross-sectoral policies for cleaner production in Iran.

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1. Introduction

Energy intensity is one of the most commonly used energy indicators in each country. In Iran, according to the IEA (2018), this

index is equivalent to 0.51 which is 15 times and 10 times higher than Japan and European Union respectively due to the high subsidized energy supply in Iran (Gudarzi Farahani and Varmazyari, 2012). But the energy consumption from the following two aspects in Iran is not considered by researchers and policymakers. The first aspect is energy consumption analysis at sectoral level and the second is the role of economic sectors in securing energy demand and growth of value added. For example based on Power Ministry of Iran (2019), although, the electricity sector has the

* Corresponding author.

E-mail addresses: ali.faridzad@atu.ac.ir (A. Faridzad), banouei7@yahoo.com (A.A. Banouei), jillianbanouei60@gmail.com (J. Banouei), zeinab.golestan@yahoo.com (Z. Golestan).

most energy intensity as compared with the other sectors which is about 25 percent, which has a small share of about 0.6 percent in supplying GDP and 1.5 percent in value added growth. Since sustaining economic growth at the sectoral level requires identification of energy intensive key sectors, measurement of the direct and indirect levels of energy consumption of each sector is very crucial for energy management policy and cleaner production in Iran.

Hence, in order to meet the energy management policy, it is very important to employ a decomposition method to identify if a sector uses energy through its own production process directly and indirectly or if it is responsible for energy consumption that other sectors generate. On the other hand, economic sectors have different weights in the economy. Economic sectors facing with a large final demand, such as household or government expenditures, have a greater impact on sectoral activity and, ultimately, energy consumption at the level of economic sectors. In the case of production factor structure, depending on capital or labor intensity of the production of each sector, the amount of energy consumption will be different. Securing these objects especially, sectoral energy management policy and recognizing that the main source of energy intensity can lead to use modern energy in production and finally secure cleaner production in sectoral level at the medium-term and long-term.

For the reasons given above, one of the innovative methods for these analyses, which simultaneously consider the sectoral structure of energy consumption and the weights of the final demand and value added, is input-output analysis. This method provide us the calculation of decomposed inter-industry multipliers, as well as the share of final demand and value added of each sector on the basis of which we can take into account the weights of the different sectors in the Economy. Piaggioab et al. (2014) demonstrated that simple Leontief demand-pull multipliers cannot indicate whether energy consumption of a sector is intensified by itself or whether it is making other sectors to demand or consume more energy.

Findings of this study based on theoretical aspects show that these characteristics of input-output model for identification of energy intensive key economic sectors regarding the two mentioned above approaches have not been considered in Iran's literature. Piaggioab et al. (2014) and Alcántara et al. (2010) extended this key sector identification model in which they study GHG emissions measurement based on decomposed multipliers of Uruguay and electricity consumption analysis of Spain respectively. In this article, input-output modern approach multipliers¹ will be categorized into two approach as decomposed weighted Leontief demand-driven and Ghosh supply-driven model which were in the following key sector concept of Ghosh (1958) and Hirshchman (1985) approaches.

In this paper, we employ decomposed multipliers with *Backward* and *Forward* linkages that can be decomposed into two components according to the weight of final demand and value

added: *Own weighted backward linkages* represent how changes in final demand of economic sector affect energy consumption from the sector itself. The second component, *pure weighted backward linkages* indicates how variations of final demand of a sector impact on energy consumption from other sectors. Also, we can represent *Own weighted forward linkages* and pure weighted forward linkages as another two components. To this end, key sector analysis is determined with multipliers with own weighted multipliers based on decomposition analysis.

In fact, based on Piaggioab et al. (2014) technical improvements and better practices are effective in energy management policies in sectors with a high own backward or forward component. Sectoral measures to reduce energy intensity are not effective in sectors with a high pure backward or forward component because these sectors are not directly responsible for energy consumption. In this case, technological or better policy making measures are effective if they reduce intermediate demand or supply to directly energy intensive sectors. As compared to the existing literature in Iran, the novelty of this articles are as follows:

- 1) Many studies in Iran, using both traditional and modern approaches, have focused on identifying key sectors of Iranian Economy (Jahangard (1998), Banouei et al. (2007), Yousefi (2013), Sadeghi and Mousavi Nik (2017)), however energy and environmental issues have not been taken into account in these analyses. As mentioned earlier, this study analyses the energy demand impact of economic activities for Iran based on a combination of Leontief and Ghosh models which leads to the decomposed weighted multipliers. The distinction of own and pure weighted backward and forward multipliers based on Alcántara et al. (2010) provide us with a way to discriminate between those sectors where policy measures should be targeted at their demand and those where instruments should rather be oriented to their production structure (Alcántara et al. (2010), 2089). While this approach has not so far been explored in Iran. For example Guo et al. (2018) identified the key sectors driving China's energy consumption and CO₂ emissions from the input-output relationship and demand elasticity. Their findings reveal that the government should pay close attention the sectors of manufacture of basic chemicals, construction of buildings, wholesale and retail trade, road transport, and real estate to achieve energy conservation target and low carbon development. Zhang et al. (2017) explores the indirect energy consumption and CO₂ emission from household consumption using the input-output method, and also the influencing factors of the indirect CO₂ emissions in China. The results show that the indirect energy consumption reaches 69–77% of the total household energy and the indirect CO₂ emission accounts for 77–84% of the total household CO₂ emissions. Per capita household consumption and energy intensity are the main influencing factors of the indirect CO₂ emission. Guerra and Sancho (2010) use an adaptation of the hypothetical extraction method to measure the role of energy and non-energy efficiency gains in an interconnected, multi sectoral Economy for Spain (Guerra and Sancho, 2010, 831). Also, Alcántara et al. (2010) identify those sectors that contribute most to electricity consumption in Spain, using a methodology based on input-output Tables. Kahrl and Ronald-Holst (2009) and Kereshner and Hubacek (2009) are another examples in which they combine input-output analysis and energy issues of economic activities.
- 2) One of the main alternative approaches for the decomposed weighted multipliers in literature is hypothetical extraction method (HEM) which is a new approach among the other modern groups. Based on HEM model additionally to the own and pure effects of multipliers, this method has residual

¹ These two basic models can be represented according to the two basic approaches for the measurement of multipliers and identification the key sectors in input-output analysis (Sánchez-Chóliz and Duarte (2003)). The first multipliers can be shown as *traditional approach* which is called unweighted backward and forward multipliers (based on Piaggioab et al. (2014)) but ignore the relative importance of sectors (Jahangard (1998) Lenzen (2003), Alcántara and Padilla (2003), Keshavarz Hadad (2004), Banouei et al. (2007)). It means this approach cannot take into account the weights of the different sectors according to final demand and value added sections of input-output Table. The second approach, as we call *modern approach*, can be categorized as several methods with different criteria such as weight of final demand and value added, Eigen Vector method (Dietzenbacher (1992)), Average Propagation Length (Dietzenbacher (2007)), Network Theory (Friedkin (1991), García Muñiz et al., (2008) and García Muñiz and Ramos (2012)), Hypothetical Extraction Method (Miller and Lahr (2001); Dietzenbacher and Lahr (2013)) and Net Multiplieres (Oosterhaven and Stelder et al. (2002)).

multipliers as mixed effect which allows the decomposed model to be balanced so that the sum of the two pure and net effects can not be equal to the gross effect without regarding the mixed effect. Miller and Blair (2009) also decomposed multipliers into net and gross effects that there is no difference mathematically with own and pure effects of our approach. Accordingly, the decomposed weighted multiplier method without generating residual multiplier as mixed effect, compared to the HEM is the more precise and preferred approach. Also, in the proposed method for this study developed by Alcántara et al. (2010) and Piaggioab et al. (2014) in contrast with HEM method, there is no need to extract or omit a sector for measurement of the effect of that sector in the entire Economy for identifying a key sector. Accordingly, it is preferable to consider all economic sectors simultaneously in the analysis for determining energy intensive key sectors. This evidence is not considered in the previously mentioned studies in justifying the use of decomposed weighted multiplier approach.

To address the policymaking questions of this paper, we have organised it as follows: The following Section presents the details of the methodology of decomposed weighted multipliers based on input-output approach and describes the database employed. Section 3, applies this method to the Iranian case and discusses the empirical results. And finally, policy implications and conclusions are discussed in Section 4.

2. Methodology and database

The purpose of this study is to analyze the energy impacts of economic activities based on identification of energy intensive key sectors of the Iranian Economy. Input-output model has been employed according to the pioneering work of Wassily Leontief in 1941. In the following paragraphs, vectors will be defined as column vectors, the transposition of matrix will be represented by ($'$) and the diagonal matrix will be indicated by ($\hat{\cdot}$). Based on Alcántara et al. (2010) and Piaggioab et al. (2014), at first we introduce the energy consumption vector ($n \times 1$) defined as matrix e , the vector of direct energy consumption which is used by the n economic sectors and let x be the vector ($n \times 1$) of production by economic sectors. The matrix W' is the vector ($1 \times n$) of energy intensities calculated as follows:

$$W' = e' \hat{x}^{-1} \quad (1)$$

The element w_i indicates the energy consumption required for the production of one unit of economic sector. We can rewrite the Eq. (1) as follows:

$$e = \hat{w}x \quad (2)$$

Substituting Leontief's demand model ($x = (I - A_d)^{-1}y$) in Eq. (2) with x , we get the following in Eq. (3):

$$e = \hat{w}(I - A_d)^{-1}y \quad (3)$$

where $(I - A_d)^{-1}$ is the inverse ($n \times n$) matrix as a domestic² production multiplier and Y represents the vector ($n \times 1$) of final demand. According to Eq. (3), $\hat{w}(I - A_d)^{-1}$ as a ($n \times n$) matrix can be simplified into W which means a W_{ij} matrix with ($n \times n$) dimension ($(W_{ij})_{(n \times n)}$). The element W_{ij} represents the transformations of final demand variations of sector j into energy consumption of sector i . The sum of column of each sector defined W_j measure the direct and indirect effect of energy intensity, per unit of final demand of sector j . Therefore, we can express this calculation for all economic sectors as Eq. (4):

$$\epsilon'_y = u'W \quad (4)$$

u is the vector ($1 \times n$) of unity and ϵ_y is the vector ($n \times 1$) of total energy intensities. This vector indicates the energy consumption multiplier effect based on per unit of final demand variation of all sectors. In the literature, this multiplier is known as *energy backward linkages*. But as presented in the previous section, these linkages are total potential multipliers and ignore the weights of the different sectors according to final demand. Ignoring the share of final demand as a weight for each sector can lead to underestimation or overestimation measurement of the energy backward linkage. Hence, we can define the vector as the share of final demand which shows the degree of each sector impact on energy consumption. Thus, $\tilde{y}$ is the ($n \times 1$) vector weighted of each sector based on final demand ($\tilde{y}_j = \frac{y_j}{\sum y_j}$). Accordingly, then $\sum \tilde{y}_j = 1$, we can indicate a new multiplier included share of final demand for each sector as weighted energy backward multiplier represented in Eq. (5).

$$\mu'_y = u'W\tilde{y} \quad (5)$$

The Eq. (5) can be indicated in terms of j th economic sector as Eq. (6):

$$\mu_{j,y} = \sum_{i=1}^n W_{ij}\tilde{y}_j \quad (6)$$

The Eq. (6) $\mu_{j,y}$ represents the total energy consumption of sector j per unit of final demand. This weighted energy backward multiplier as $\mu_{j,y}$ can be decomposed into two types of backward multiplier. The first component is *pure weighted backward multiplier* when $i \neq j$ which indicates how variations of weighted final demand of sector j impact on energy consumption from sector i . This component can be measured mathematically as Eq. (7):

$$\mu_{ij,y}^{BP} = \sum_{i \neq j} W_{ij}\tilde{y}_j \quad (7)$$

Also, the other component of decomposed weighted backward multiplier is *own weighted backward multiplier* which is represented as Eq. (8):

$$\mu_{jj,y}^{BO} = W_{jj}\tilde{y}_j \quad (8)$$

The *own weighted backward multiplier* denotes that how changes in weighted final demand of sector j affect energy consumption from the own sector (sector j).

Similarly, supply driven multipliers defined by Ghosh (1958) and introduced by Jones (1976) have been used. According to the supply driven multipliers, instead of using the Leontief technical coefficients, the distribution coefficient matrix is employed which leads to the Ghosh inverse matrix as Eq. (9):

² All the Input-Output Tables which have so far been estimated and compiled in Iran, are based on the assumption of competitive imports. We have used the method suggested by Pei, J. et al. (2011) for separation imports from intermediate matrix which is explained in database section of this article. Respectively, the technical coefficient of Leontief multiplier is measured domestically based on assumption of competitive imports and necessity of import separation. In addition, as we refer to the Faridzad and Banouei (2017) in section 2, for applying domestic production multiplier in which it is important to measure the actual size of multiplier without any underestimation.

Table 1
Classification of normalized multipliers.

Supply side multipliers (Normalized Weighted Forward Linkages)		
Demand side multipliers (Normalized Weighted Backward Linkages)	$\mu_{i,v}^n > 1$	$\mu_{i,v}^n < 1$
	$\mu_{j,y}^n > 1$	Energy intensive key sectors
	$\mu_{j,y}^n < 1$	supply push energy intensive sectors
		demand push energy intensive sectors
		Rest of Sectors

Source: Own elaboration and extension from Alcántara et al. (2010).

$$(I - B_d)^{-1} = \widehat{x}^{-1} (I - A_d) \widehat{x} \tag{9}$$

where B_d is the domestic distribution coefficient matrix with $(n \times n)$ dimension in which B_{ij} represents the share of the production of sector i employed in the production of sector j . So the matrix Z can be defined with $(n \times n)$ dimension as Eq. (10) in which Z_{ij} indicates energy ecnsumption in production of sector j per unit of sector i produced.

$$Z = (I - B_d)^{-1} \widehat{W} = (Z_{ij})_{n \times n} \tag{10}$$

Also, in Eq. (10) the matrix W based on W_{ij} element represents the transformation of value-added variations of sector i into energy consumption of sector j . The sum of any row i of the matrix Z can lead to the direct and indirect effects on energy consumption due to the production of a unit of sector i known as *energy forward multipliers*. Hence, we can write this calculation for all economic sectors as Eq. (11):

$$\varepsilon_v = Zu \tag{11}$$

These linkages are potential multipliers as energy backward multipliers mentioned before. Ignoring the share of value added as a weight for each sector can be lead to higher or lower of the energy forward linkage measurement. Hence, we can define the vector as the share of value added which indicates the share of each sector impact on energy consumption. Thus, $\widehat{v}$ is the $(1 \times n)$ vector weighted of each sector's value added ($\widehat{v}_i = \frac{v_i}{\sum v_i}$). Accordingly, then $\sum v_i = 1$, it is indicated a new multiplier included share of value added for each sector as weighted forward multiplier represented in Eq. (12).

$$\mu_v = \widehat{v}Zu \tag{12}$$

Similar to the demand driven approach, μ_v is the total energy consumed by each sector per unit produced which can be decomposed into two components according to weighted energy forward multipliers. The first component as $\mu_{i,v}^{FP}$ is pure weighted forward multiplier represented as Eq. (13):

$$\mu_{i,v}^{FP} = \sum_{i \neq j} Z_{ij} \widehat{v}_i \tag{13}$$

This component indicates how variations in the production of sector i impact on energy consumption of the other sectors (sector j). Also, the other component of decomposed weighted forward multiplier is *own weighted forward multiplier* as $\mu_{i,v}^{FO}$ which is represented as Eq. (14):

$$\mu_{i,v}^{FO} = Z_{ii} \widehat{v}_i \tag{14}$$

The *own weighted forward multiplier* denoted that how variations in the production of economic sector i affect energy consumption from the own sector (sector i).

Since, according to literature review, the weighted multipliers determine the energy intensive key sectors for the Economy, we need an indicator for distinguishing these key sectors based on the

four identified types of linkages. This indicator can be defined as “Normalized Average Multipliers” according to both demand and supply perspective in this paper. This average proposed by Alcántara et al. (2010) is equivalent for both approaches and can be calculated as Eq. (15):

$$\mu^n = \frac{n \sum_{i=1}^n \mu_{i,v} u}{\sum_{i=1}^n \sum_{j=1}^n \mu_{i,v} u} = \frac{n \sum_{j=1}^n \mu_{j,y} u}{\sum_{i=1}^n \sum_{j=1}^n \mu_{j,y} u} = 1 \tag{15}$$

Accordingly, we extend the decomposed linkages to normalized average multipliers which are presented in four classifications as Table 1.

According to Table 1, energy intensive key sectors can be identified from both the supply and demand perspectives and, finally, the determined policymaking priorities. Considering the amount of decomposed multipliers and situation of sectors in Table 1, we can identify the amount of energy consumption of sectors due to own linkages or pure linkages.

For studying energy impacts of economic sectors, we employ a symmetric Iran input-output 71×71 Table (industry-by-industry) for the year 2011 which is provided by *Statistical Center of Iran* (Statistical Center of Iran, 2019a,b) This Table is the most up to date and the last Table in Iran released in 2015 which is aggregated to a symmetric input-output 26×26 Table due to the limitation of primary energy data categorized in each sector. The information on primary energy consumption of economic sectors is obtained from Energy Balance Sheet of Iran provided by *Power Ministry of Iran* for the year 2011 (Power Ministry of Iran, 2019). The energy consumption data in Hydrocarbon Balance Sheet classified into 9 general sectors should be extended to the 26 economic sectors according to the *International Standard Industrial Classification of All Economic Activities (ISIC)*, Revision 4 (Statistical Center of Iran, 2019a,b) as shown in Table 2³. These adjustments are made for input-output analysis.

According to the Table 2, fossil energy consumption of Iranian economy for 26 sectors is observed in energy intensive sectors especially industrial sub sectors at the first glance. It is important to note that this information of energy consumption is measured direct energy consumption of each sector. Based on Table 2, it is noticeable that the production and distribution of electricity has the most energy consumption across the other sectors, however in contrast it has the least final demand share. The two other economic sectors, transportation and crude oil and natural gas extraction are in the next rank but have the large share of final demand and value added other than electricity sector. As it was expected, the main other economic sectors in which they have a large direct energy consumption are manufacturing sectors, meanwhile some sectors such as services in the 7th rank and agriculture and other mining sectors in the 11th rank of direct energy consumption show the high level of energy consumption

³ For the extension of energy consumption data of 26 economic sectors, due to the limitation of energy consumption data for economic sectors (Just for 9 sectors), Authors prepared and calculated theses data manually according to the *International Standard Industrial Classification of All Economic Activities (ISIC)*, Revision 4.

Table 2Fossil Energy Consumption⁴/Final Demand Share/Value Added Share of Iranian economic sectors in 2011 (Unit for Energy Consumption: Barrels of Oil Equivalent).

Rank of Sector	Sector	Energy Consumption	Final Demand Share (%)	Value Added Share (%)
1	Production and distribution of Electricity	408,768,968	0.06	1.5
2	Transportation	302,661,590	4.7	5.6
3	Crude oil and natural gas extraction	227,349,411	14.3	16.1
4	Manufacture of chemicals and chemical products	202,780,000	2.6	2.1
5	Manufacture of other non-metallic mineral products	158,824,297	0.5	1.2
6	Manufacture of basic metals	78,868,575	0.6	0.6
7	Services	57,879,667	42.2	42.9
8	Manufacture of coke and refined petroleum products	52,052,000	3.5	3.6
9	Manufacture of food products and beverages	38,120,185	4.5	1.7
10	Manufacture of gas; distribution of gaseous fuels	36,927,348	2.1	5.2
11	Agriculture and Other mining and quarrying	35,552,887	8.1	8.9
12	Manufacture of wood, textiles, paper products	10,948,806	0.5	0.6
13	Manufacture of motor vehicles and trailers	6,458,795	2.6	1.2
14	Manufacture of fabricated metal products	4,356,063	0.6	0.6
15	Manufacture of rubber and plastics products	3,811,201	0.0	0.3
16	Repair and installation of machinery and Equip.	3,626,024	0.9	0.5
17	Manufacture of furniture	3,306,706	0.9	0.4
18	Manufacture of electrical equipment	1,607,022	0.3	0.3
19	Manufacture of other transport equipment	556,070	0.2	0.1
20	Printing and reproduction of recorded media	455,623	0.0	0.1
21	Water collection, treatment and supply	432,503	0.1	0.3
22	Manufacture of pharmaceuticals products	386,169	0.1	0.1
23	Construction activities	217,830	9.6	5.2
24	Motion picture, video and television production,	95,435	0.1	0.0
25	Manufacture of tobacco products	84,259	0.1	0.0
26	Manufacture of computer and optical products	54,771	0.1	0.0
Sum		1,636,182,204	100	100

Source: Prepared by the authors' calculation based on Energy Balance sheet of Iran (2011) and Input-Output Table of 2011.

between the manufacturing sectors. For instance, the services has the most share of final demand and value added of Iranian economy with the share of more than 42 percent. Hence, it was not surprisingly that the direct energy consumption of services is given the 7th-place rank between the other economic sectors.

In addition, it is necessary to employ domestic inter-sectoral intermediate matrix for generating the domestic coefficient matrix or the distribution coefficient matrix which is extended by Banouei (2011). Import deduction as a methodology based on competitive assumption⁵ generates domestic inter-sectoral intermediate matrix. Employing the input-output Table regardless to the import deduction lead to underestimation of actual multipliers (Faridzad and Banouei (2017)) and it also inappropriate for measuring a correct quantitative evaluation of ecological footprints of natural resources and international trade (Bicknell et al. (1998), Freng (2002) and Freng (2001)). There are three methods for import deduction based on Pie et al., 2011 and Banouei (2011), however it has used the 3rd method for deducting import in this study. Hence, from the technical point of view, the database is ready to be employed for identifying the energy intensive key sectors of Iran based on a structural relationship between economic activities and fossil energy consumption. These factors are addressed in the next section.

3. Empirical results

In this section, weighted backward and forward multipliers are decomposed into two components for determining the energy intensive key sectors of Iran based on the normalized average multipliers shown in Table 1.

As shown in Table 3 and Table 4, the results of own weighted backward linkages and pure weighted backward linkages are presented respectively. As mentioned in the last sections, the own weighted backward linkages (expressed as OWBM in Table 3) shows how changes in final demand of economic sector *j* affect energy consumption from the own sector (sector *j*). The results in Table 4 reveals that crude oil and natural gas extraction, transportation and electricity production and distribution are the first three sectors which have the most energy consumption due to their demand of their own sectors.

For example, it is observed that increase in one unit of weighted final demand, the electricity sector would require 23.177 barrels of oil equivalent fossil energy for electricity production and distribution. In addition, according to the Table 3, the service sector, due to its high share in the Iranian Economy (about 42.2 percent), mainly because of wholesale and retail sector (about 50 percent of service sector), has a higher energy consumption compared to many other manufacturing sectors such as manufacture of other non-metallic mineral products and manufacture of coke and refined petroleum products. Also, since the crude oil and natural gas extraction and transportation sectors have high share of final demand (14.3 and 4.7 respectively), it is expected that the own weighted backward linkage has a larger multiplier.

The last column of Table 3 illustrates the normalized amount of OWBM of each sector which indicates that each sector has a normal amount more than unity, and has more energy intensity compared to the average of the Economy (this average is unit for the whole of the Economy). Hence, for instance, the normalized indicator for transportation sector is 5.823 which show that energy intensity of transportation sector by increasing its weighted final demand is

⁴ About the fossil energy consumption data, it is very important to note that the other modern non-fossil energy consumption applying in measurement of energy intensive key sectors of Iranian economic, according to the statistical data of power ministry of Iran (2019), the share of these mentioned type of energy has the less than 5 percent in energy consumption in the level of Iranian economic sectors as intermediate goods. Hence, it can be negligible for final calculations.

⁵ All the Input-Output Tables which have so far been estimated and compiled in Iran, are based on the assumption of competitive imports. The authors use the method suggested by Pei, J. et al. (2011).

Table 3

Own weighted backward Multipliers
For 26 Iranian Energy Intensive Sectors in 2011.

Rank of Sector	Sector	OWBM ^b	OWBM ^{nc}
1	Crude oil and natural gas extraction	31.711	6.158
2	Transportation	29.984	5.823
3	Production and distribution of Electricity	23.177	4.501
4	Manufacture of chemicals and chemical products	15.100	2.932
5	Services ^a	7.888	1.532
6	Manufacture of other non-metallic mineral products	5.755	1.118
7	Manufacture of food products and beverages	4.748	0.922
8	Manufacture of coke and refined petroleum products	3.991	0.775
9	Agriculture and Other mining and quarrying	3.622	0.703
10	Manufacture of basic metals	2.574	0.500
11	Manufacture of gas; distribution of gaseous fuels	2.235	0.434
12	Manufacture of motor vehicles, trailers and semi-trailers	0.922	0.179
13	Manufacture of wood, textiles, paper products	0.740	0.144
14	Manufacture of furniture	0.430	0.084
15	Repair and installation of machinery and equipment	0.417	0.081
16	Manufacture of fabricated metal products	0.274	0.053
17	Manufacture of electrical equipment	0.106	0.021
18	Manufacture of other transport equipment	0.074	0.014
19	Manufacture of pharmaceuticals products	0.047	0.009
20	Construction activities	0.029	0.006
21	Water collection, treatment and supply	0.023	0.004
22	Motion picture, video and television production,	0.013	0.002
23	Manufacture of tobacco products	0.012	0.002
24	Manufacture of computer and optical products	0.007	0.001
25	Printing and reproduction of recorded media	0.005	0.001
26	Manufacture of rubber and plastics products	0.002	0.000

^a Rest of services includes wholesale trade, retail trade, Telecommunication services, real-estate services, public administration.

^b Own Weighted Backward Multipliers.

^c Normalized Own Weighted Backward Multipliers.

Source: Prepared by the authors based on Eq. (8)

about 5.8 times higher than the average of the Economy. This indicator for crude oil and natural gas extraction has taken the most value which is about 6.15 times higher than the average of the

Economy.

In addition, these normalized OWB multipliers illustrate another vision of reality. The sectors whose normalized value are

Table 4

Pure weighted backward Multipliers
For 26 Iranian Energy Intensive Sectors in 2011.

Rank of Sector	Sector	PWBM ^b	PWBM ^{nc}
1	Services ^a	37.212	8.871
2	Construction activities	28.517	6.798
3	Agriculture and other mining and quarrying	9.156	2.182
4	Manufacture of food products and beverages	6.837	1.63
5	Manufacture of coke and refined petroleum products	4.042	0.964
6	Transportation	3.266	0.779
7	Manufacture of chemicals and chemical products	3.208	0.765
8	Manufacture of motor vehicles, trailers and semi-trailers	3.185	0.759
9	Manufacture of furniture	2.151	0.513
10	Repair and installation of machinery and equipment	1.751	0.417
11	Manufacture of basic metals	1.589	0.379
12	Manufacture of fabricated metal products	1.543	0.368
13	Crude oil and natural gas extraction	1.471	0.351
14	Manufacture of other non-metallic mineral products	1.430	0.341
15	Manufacture of wood, textiles, paper products	1.009	0.240
16	Manufacture of computer and optical products	0.763	0.182
17	Water collection, treatment and supply	0.572	0.136
18	Manufacture of gas; distribution of gaseous fuels	0.442	0.105
19	Manufacture of other transport equipment	0.309	0.074
20	Manufacture of pharmaceuticals products	0.232	0.055
21	Production and distribution of Electricity	0.167	0.040
22	Manufacture of electrical equipment	0.102	0.024
23	Manufacture of tobacco products	0.048	0.012
24	Motion picture, video and television production,	0.044	0.011
25	Printing and reproduction of recorded media	0.014	0.003
26	Manufacture of rubber and plastics products	0.012	0.003

^a Services include wholesale trade, retail trade, Telecommunication services, real-state services, public administration.

^b Pure Weighted Backward Multipliers.

^c Normalized Pure Weighted Backward Multipliers.

Source: Prepared by the authors based on Eq. (7)

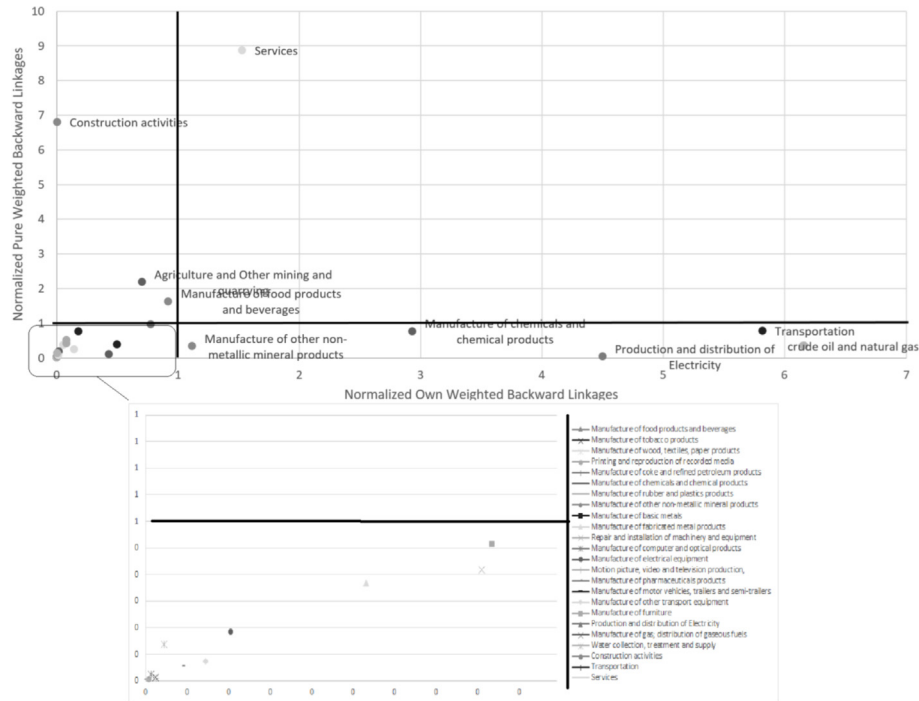


Fig. 1. Iranian Energy Intensive Key Sectors based on Demand Impacts
Source: Derived from Eq. (7) and Eq. (8).

more than unity, are energy intensive sectors such as the sectors ranked from 1 up to 6, while the others in spite of positive values act lower than the average of the Economy and have a lower energy intensity. These sectors based on Table 3, ranked from 7 up to 26.

According to Table 4, pure weighted backward multipliers measured according to the Eq. (7), indicates how variations of final demand of sector j impact on energy consumption from the other economic sectors (sector i). According to Table 4, Iranian sectors ranked for 26 sectors in which the services, construction activities and agriculture and other mining sectors are the first three sectors with the most value of pure weighted backward linkages.

The value of pure weighted multiplier for each sector indicates that one unit of weighted final demand increase in construction sector requires 28.517 barrels of oil equivalent as fossil energy for the production the other 25 sectors. The similar interpretation can be used for other sectors. The results also show that the services and construction sector have a high final demand share in Iran's Economy as mentioned in Table 2. The share of final demand for two above mentioned sectors are 42.2 and 9.6 percent respectively. And as compared to the other sectors, it is extraordinary high. This higher final demand share in the Economy is expected to be higher weighted backward linkages and more energy intensity for the other economic sectors through more supply of goods and services to sector j .

In addition, the normalized indicator in the last column of Table 4 shows that, for example, service sector due to one unit increase in its final demand, has 8.87 times higher energy intensity than the average of the economy which suggests to more direct and indirect energy consumption of the other economic sectors. On the other hand, the sectors, whose ranks from 5 up to 26, have lower energy intensity compared to the top 5 sectors which appears lower than the average of economy.

Fig. 1 plots the normalized own and pure weighted backward multipliers for showing the importance of economic sectors in energy consumption from the demand impact perspective. The

horizontal and vertical thick black line indicate the average value of own and pure weighted backward linkages measured by Eq. (15) in which after normalization, this value is equal to unit for own and pure values. Fig. 1 can be divided in four groups. First group shows the sectors with low own weighted backward multipliers but high pure weighted multipliers. Second group have high own and pure weighted backward multipliers. The third group shows the sectors with high own weighted backward multipliers but low pure weighted multipliers and the last group reveals the sectors with low own and pure weighted backward multipliers.

For example, the production and distribution of electricity is one of the sectors in the third group. It means that the electricity sector had a strong impact from the variation of final demand which leads to strong demand of fossil energy by the electricity sector itself (sector i) but the variation of final demand of the other economic sectors (sector j) cannot influence fossil energy consumption in the electricity sector significantly. This shows that the sectors such as electricity, transportation and crude oil and natural gas extraction with high OWB and low PWB multipliers are high energy intensity in fossil energy consumption.

For example, in electricity sector of Iran, some electricity generated at each power plant is utilized by the equipment and machinery of the same power plant. Therefore, the electricity delivered to the transmission networks as the output of the power plants is less than the value of the measuring devices of the generators. The share of self-consumption of electricity in power plants was 3.5% and the electricity distribution network losses and electricity transmission losses is 15.47 and 4.79 million barrels of oil equivalent in 2015, based on Power Ministry of Iran (2019). Hence, it is expected that the electricity sector due to these losses has high own weighted effect in Iran.

In addition, it can explain the situation of other economic sectors according to the Fig. 1.

For example, construction is in the first group. This sector accompanies high pure but low own weighted backward linkages

indicated that variation of final demand of the other economic sectors (sector j) can significantly affect fossil energy consumption in the construction sector. So the energy intensity of the construction sector should be controlled. However, urban buildings and the other development activities like construction roads and bridges in Iran are not fossil energy intensive, according to Table 2, reducing the final demand of the products of the other economic sectors would reduce the fossil energy consumption in construction sector. Specially, based on Table 2, final demand of construction sector with share of 9.6% occurs from energy intensive sectors such as non-metallic mineral products produce. Also, the sector such as services with high pure and own weighted backward multipliers in second group is considerably influenced by the variation of final demand made from all dependent economic sectors and leads to more energy consumption. In contrast, for example, the sectors specially, manufacture of rubber and plastic products which are in the fourth group, cannot significantly impact fossil energy consumption through the variation of final demand of itself and the other sectors.

On the other hand, the supply impacts on the energy intensity of key sectors of the Iranian Economy can be considered in Table 5 and Table 6. According to Table 5, the energy multipliers of supply driven Ghosh approach can be decomposed to two multipliers as own weighted forward multipliers and pure weighted forward linkages for determining the energy intensive key sectors of Iran based on the criteria illustrated in Table 1.

Regarding Table 6, as mentioned in the last section, the own weighted forward linkages (expressed as OWFM in Table 5) shows that how variations in the production of sector i affect energy consumption from the own sector (sector i). According to Table 5, Iranian sectors such as electricity production and distribution, the crude oil and natural gas extraction and transportation ranked the first three sectors have the most energy consumption due to the

variations in the production of their own sectors respectively.

It is shown, for example that with increase in one unit of weighted value added, the production and distribution of electricity sector for its production would require 54.837 barrels of oil equivalent fossil energy. Also, according to Table 3, since the crude oil and natural gas extraction and transportation sectors have high share of value added in Iranian Economy (16.1 and 5.6 respectively), it is expected that the own weighted forward linkage has a larger multiplier.

The last column of Table 5 presents the normalized amount of OWFM of each sector which indicates that each sector that has a normal amount more than unity has more energy intensity compared to the average of the Economy (this average is the unit for the whole of Economy). Hence, for instance, the normalized indicator for transportation sector of 5.041 indicates that energy intensity of transportation sector by increasing its value added is about 5 times higher than the average of the Economy. This indicator for electricity has taken the most value which is about 7.82 times higher than the average of the Economy.

In addition, these normalized OWF multipliers, similar to the normalized backward linkages, represent another view of the Iranian Economy. The sectors, whose normalized values are more than unity, are energy intensive sectors such as the sectors ranked from 1 to 6, while the others in spite of positive values have energy consumption lower than the average of the Economy. These sectors based on Table 6, ranked from 7 up to 26. By comparing the results of Table 5 with Table 3, we find that the rank of the sectors according to OWFM is almost similar to OWBM.

Regarding Table 6, pure weighted forward multipliers measured according to the Eq. (13), indicates how variations of value added of sector j impact on energy consumption from the other economic sectors (sector i). In Table 6, 26 sectors are ranked in which the manufacture of gas; distribution of gaseous fuels, Services and

Table 5
Own weighted forward Multipliers
For 26 Iranian Energy Intensive Sectors in 2011.

Rank of Sector	Sector	OWFM ^b	OWFM ^{nc}
1	Production and distribution of Electricity	54.837	7.829
2	Crude oil and natural gas extraction	35.699	5.097
3	Transportation	35.308	5.041
4	Manufacture of other non-metallic mineral products	12.806	1.828
5	Manufacture of chemicals and chemical products	11.958	1.707
6	Services ^a	8.021	1.145
7	Manufacture of gas; distribution of gaseous fuels	5.615	0.802
8	Manufacture of basic metals	5.488	0.784
9	Manufacture of coke and refined petroleum products	4.114	0.587
10	Agriculture and Other mining and quarrying	3.967	0.566
11	Manufacture of food products and beverages	1.804	0.258
12	Manufacture of wood, textiles, paper products	0.895	0.128
13	Manufacture of motor vehicles, trailers and semi-trailers	0.438	0.063
14	Manufacture of fabricated metal products	0.278	0.040
15	Repair and installation of machinery and equipment	0.214	0.031
16	Manufacture of furniture	0.211	0.030
17	Manufacture of rubber and plastics products	0.197	0.028
18	Manufacture of electrical equipment	0.101	0.014
19	Water collection, treatment and supply	0.043	0.006
20	Printing and reproduction of recorded media	0.037	0.005
21	Manufacture of other transport equipment	0.026	0.004
22	Manufacture of pharmaceuticals products	0.025	0.004
23	Construction activities	0.016	0.002
24	Manufacture of tobacco products	0.007	0.001
25	Motion picture, video and television production,	0.007	0.001
26	Manufacture of computer and optical products	0.003	0.000

^a Rest of services includes wholesale trade, retail trade, Telecommunication services, real-state services, public administration.

^b Own Weighted Forward Multipliers.

^c Normalized Own Weighted Forward Multipliers.

Source: Prepared by the authors based on Eq. (14)

Table 6

Pure weighted forward Multipliers
For 26 Iranian Energy Intensive Sectors in 2011.

Rank of Sector	Sector	PWFM ^b	PWFM ^{bc}
1	Manufacture of gas; distribution of gaseous fuels	6.006	13.439
2	Services ^a	5.362	11.999
3	Manufacture of coke and refined petroleum products	3.407	7.623
4	Agriculture and Other mining and quarrying	3.225	7.216
5	crude oil and natural gas extraction	1.359	3.041
6	Production and distribution of Electricity	1.159	2.594
7	Transportation	0.906	2.028
8	Manufacture of chemicals and chemical products	0.891	1.994
9	Construction activities	0.623	1.394
10	Manufacture of basic metals	0.61	1.366
11	Manufacture of electrical equipment	0.426	0.953
12	Manufacture of rubber and plastics products	0.4	0.894
13	Manufacture of wood, textiles, paper products	0.375	0.84
14	Manufacture of fabricated metal products	0.316	0.708
15	Manufacture of other non-metallic mineral products	0.244	0.546
16	Water collection, treatment and supply	0.227	0.508
17	Manufacture of food products and beverages	0.179	0.400
18	Repair and installation of machinery and equipment	0.152	0.340
19	Manufacture of motor vehicles, trailers and semi-trailers	0.039	0.088
20	Manufacture of furniture	0.037	0.082
21	Printing and reproduction of recorded media	0.025	0.056
22	Motion picture, video and television production,	0.013	0.028
23	Manufacture of other transport equipment	0.008	0.017
24	Manufacture of pharmaceuticals products	0.007	0.016
25	Manufacture of computer and optical products	0.005	0.012
26	Manufacture of tobacco products	0.000	0.000

^a Rest of services includes wholesale trade, retail trade, Telecommunication services, real-estate services, public administration.

^b Pure Weighted Forward Multipliers.

^c Normalized Pure Weighted Forward Multipliers.

Source: Prepared by the authors based on Eq. (13)

manufacture of coke and refined petroleum products are the first three sectors with the most value of pure weighted forward linkages.

It describes the value of pure weighted forward multiplier for each of the above mentioned sector. For example, an increase in one unit of weighted value added of service sector requires 5.362 barrels of oil equivalent as fossil energy for the other 25 sectors. These explanations can be applied for the other economic sectors as shown in Table 6. In addition, the manufacture of gas, distribution of gaseous fuels and manufacture of coke and refined petroleum products has a high value added share in Iran's Economy as mentioned in Table 3. The share of value added for the two above mentioned sectors are 5.2 and 3.6 percent respectively and as compared to the other sectors, it is relatively high.

In addition, the normalized indicator in the last column of Table 6 shows that, for example, service sector due to one unit increase in its value added, has 11.99 times higher energy intensity than the average of the Economy which leads to more direct and indirect energy consumption of the other sectors. The sectors, whose ranks are from 11 up to 26, have lower energy intensity compared to the top 10 sectors in which they have lower than the average of Economy in energy matters.

Also, Fig. 2 shows the normalized own and pure weighted forward multipliers which consider the importance of economic sectors in energy consumption from a supply impact perspective. The horizontal and vertical thick black line indicates the average value of own and pure weighted forward linkages measured by Eq. (15) the value of which, after normalization, is equal to unity for own and pure values. Fig. 2 similar to the Fig. 1 can be categorized into four groups. First group indicates the sectors with low own weighted forward multipliers but high pure weighted forward multipliers. The second group has high own and pure weighted forward multipliers. The third group denotes the sectors with high

own weighted forward multipliers but low pure weighted multipliers and the last group reveals the sectors with low own and pure weighted forward multipliers.

For example, the production and distribution of electricity is one of the sectors in the second group. It means that the electricity sector has strong impact from the variation of value added which leads to strong supply of goods and services to the electricity sector itself (sector i) and the other economic sectors (sector j) which has an influence on fossil energy consumption in the electricity sector and the other economic sectors significantly. This shows that the sectors such as electricity and crude oil and natural gas extraction with high OWF and PWF multipliers lead to high fossil energy consumption through their own production as compared with the other sectors.

We can also explain the situation of other sectors according to the plot division of Fig. 2.

For instance, the results in Table 6 show that the manufacture of coke and refined petroleum products is in the first group. Here, high pure is accompanied by low own weighted forward linkages. Variation of production of the other sectors (sector j) can significantly affect fossil energy intensity in the manufacture of coke and refined petroleum products. So the energy intensity of the coke and refined petroleum products should be controlled. However, manufacture of coke and refined petroleum products has high pure forward linkage which means that one unit of production of other sectors requires 3.407 barrels of oil equivalent fossil energy consumption of this sector (direct and indirect). In addition, the fossil energy loss of refineries is 7.81 million barrels of oil equivalent (1.24% loss for refinery) in 2015. Hence, with a rule of thumb calculation, for increasing one unit of production of other sectors, we can expect about 0.042 barrel of oil equivalent fossil energy loss in manufacture of coke and refined petroleum products. For instance, if the production of the economic sectors will be increased

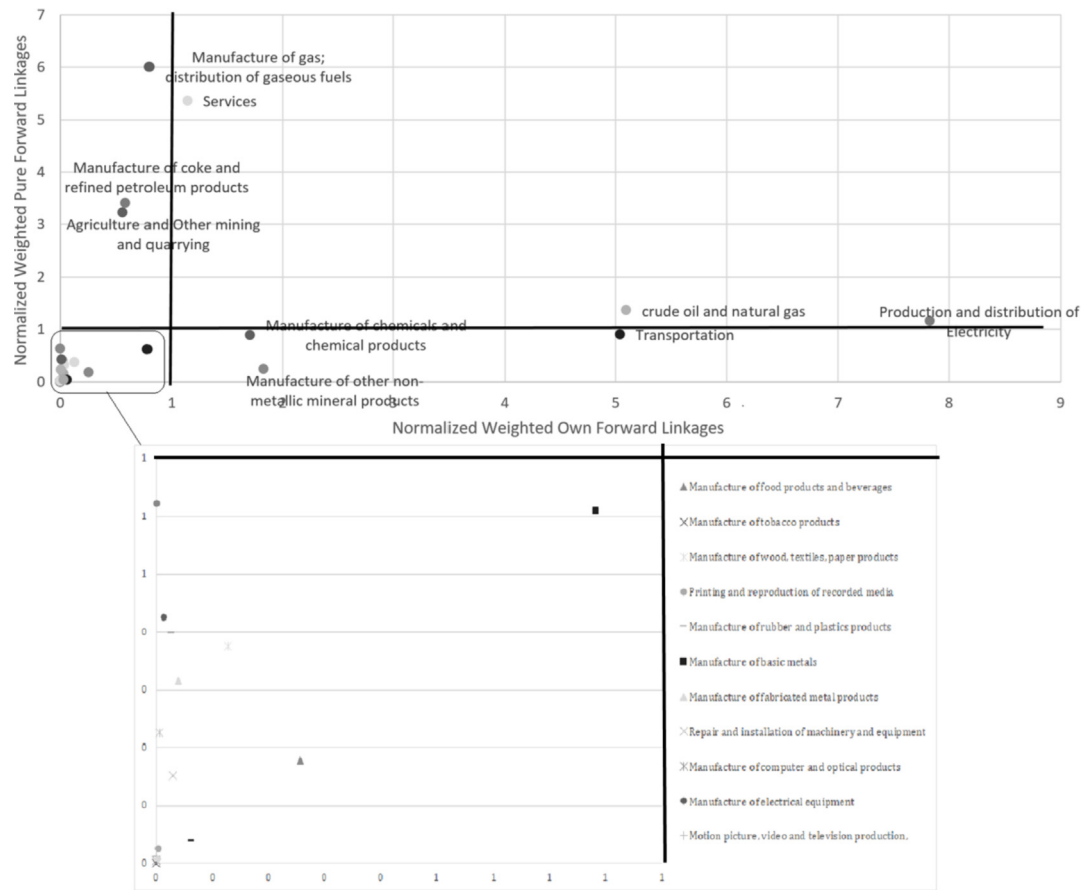


Fig. 2. Iranian Energy Intensive Key Sectors based on Supply Impacts
Source: Derived from Eq. (8) and Eq. (14).

about ten million rials in one year, the fossil energy loss of manufacture of coke and refined petroleum products is increased about 420,000 barrel of oil equivalent. The energy intensity of the manufacture of coke and refined petroleum products based on normalized amounts of PWF multipliers in Table 6, is about 7 times that of the manufacture of basic metals.

Also, the sector such as transportation with low pure and high own weighted forward multipliers in third group considerably influenced by the variation of intra industry of production and leads to more fossil energy consumption. In contrast, for example, the sectors, specially, Manufacture of computer and optical products which is in the fourth group cannot significantly impact fossil

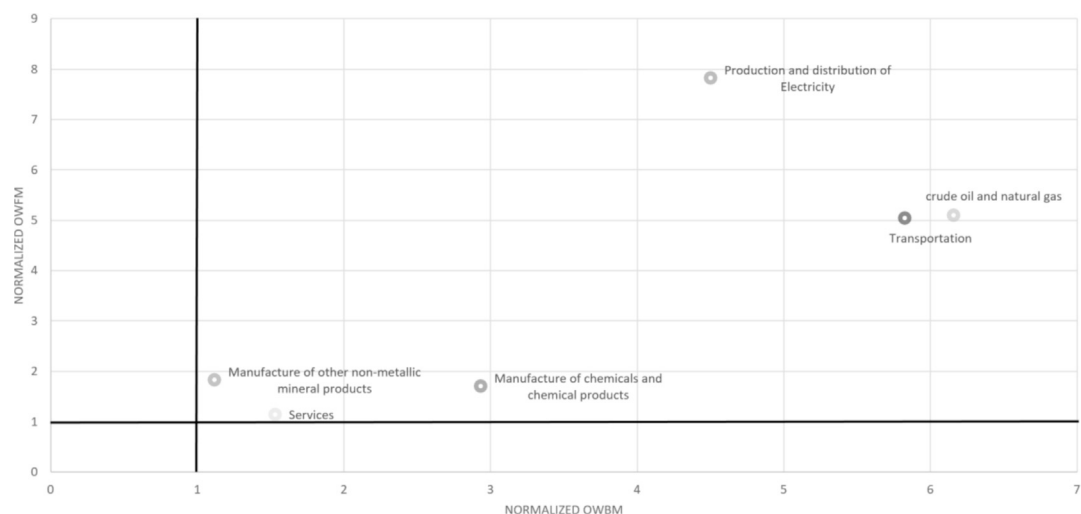


Fig. 3. Iranian energy intensive key Sectors
Source: Derived from Eq. (12) and Eq. (19).

energy consumption through the variation of production of itself and the other economic sectors. The Manufacture of computer and optical products is a high tech industry category which it is not expected to be an energy intensive industry specially based on supply impacts.

According to the results of this study, the energy intensive key sectors can be identified from Fig. 3, show the average of energy intensity of economic activities that they have more energy intensity than the other economic sectors.

As shown in Fig. 3, based on the main results of this study, six economic sectors, including, production and distribution of electricity, crude oil and natural gas extraction, transportation, manufacture of chemicals and chemical products, services and manufacture of other non-metallic mineral products are energy intensive key sectors of Iran which have the most value of own weighted backward and forward multipliers comparing to the other economic sectors. These energy intensive economic key sectors are important for Iranian policymakers as energy consumption matters.

From an energy consumption perspective, these six economic sectors among the 26 sectors studied in this paper, have the most direct fossil energy consumption share. The electricity sector with 25% of direct energy consumption is the most energy intensive sector in Iran. This sector also has the most loss in conversion and self-consumption in Iran which is about 307.79 barrels of oil equivalent in 2015 (Energy Balance sheet of Iran (2015)). Table 3, also shows that after the electricity sector, the direct fossil energy consumption share for transportation, crude oil and natural gas extraction, manufacture of chemicals and chemical products, manufacture of other non-metallic mineral products and services is about 18%, 14%, 12%, 10% and 4% respectively. In the other words, these six mentioned activities through demand and supply impact can make the most effect on indirect energy consumption due to the production of themselves and the other economic sectors.

4. Policy implications and conclusions

Regarding the evidence of this study, identifying energy intensive key sectors as a measure can help to reduce energy consumption. However, one of the reasons for the high energy intensity at the level of the economic sectors of Iran is the low energy prices that should be taken into consideration. Other recommendations to policymakers are considered under non-price policies to reduce energy intensity at energy intensive key sector levels for cleaner production. Hence, the sectors with high own weighted backward linkages in which the production of the sector lead to the more energy consumption in comparison with other economic sectors, the policy maker should manage and reduce energy consumption through substituting and adopting low energy intensive technologies. In order to meet the energy management policy, it is very important to identify if a sector uses energy

through its own production process directly and indirectly or if it is responsible for energy consumption that other sectors generate. Crude oil and natural gas extraction, transportation, production and distribution of electricity, manufacture of chemical products, services and manufacture of mineral non-metallic products with high own weighted backward multipliers through demand impacts have the most energy consumption. Also, the sectors with high own weighted forward linkages which are similar to the own backward effects have the most energy consumption through supply impacts. As shown in Table 7, these economic sectors have the highest value of own weighted backward and forward effects. The identification of the theses economic sectors according to the first purpose of this study help the policymakers to pursue the second purpose of this study for achieving lower energy intensity target and the cleaner production. Also, it is interestingly the identified energy-intensive key sectors have the highest share of the Iranian economy in 2011 with more than 66% of GDP. However, the pure weighted multipliers show that the energy intensity of economic sectors with high pure effects are influenced by the production of the other economic sectors, the policymakers for energy management should be interested predominantly to have policy on the economic sectors with high own weighted effects. Accordingly, the results of Table 7 show that these 6 sectors have the first priorities to reduce their energy intensity by energy management as the main purpose of this study for cleaner production from the policy making point of view.

According to the results of Table 7, production and distribution of electricity is one of the main energy intensive key sectors that requires implementing energy management policy. This energy management policy can be implemented by identifying the technical and technological problems and best practical solutions for cleaner production. Since the importance of production and distribution of electricity is a strategic sector, the policymaker should cautiously decide on increasing electricity production. For example, combined cycle power plants have lower internal energy consumption than other power plants due to higher efficiency compared to gas turbines, diesel-fired, or wind power plants. Hence, converting natural gas-based utilities into combined cycle ones, in addition to increasing efficiency, conserve fuel consumption in electricity production. So that each unit of steam in combined cycle power plants, on average, conserve fuel consumption by 200 million cubic meters of natural gas annually in Iran. According to the Power Ministry of Iran (2019), the energy losses in the Iranian electricity sector in 2011 are 285.43 million barrels of oil equivalent per year, which accounts for an overall loss of energy of 59.76 percent. Unfortunately, this amount of energy losses in electricity sector has reached to the 65.3% in the year 2015. Therefore, it is recommended that enhancing the technology of power plants make the efficiency from 34% to higher values, such as 55% through combined cycle power plants.

Crude oil and natural gas extraction is another energy intensive

Table 7
Iranian Energy Intensive key Sectors in 2011.

Sector	OWBM ^a	OWFM ^b	Final Demand Share (%)	Value Added Share (%)
Production and distribution of Electricity	4.501	7.829	1.5	0.06
crude oil and natural gas extraction	6.158	5.097	14.3	16.1
Transportation	5.823	5.041	5.6	4.7
Manufacture of other non-metallic mineral products	1.118	1.828	0.5	1.2
Manufacture of chemicals and chemical products	2.932	1.707	2.6	2.1
Services	1.532	1.145	42.2	42.9

^a Normalized Own Weighted Backward Multipliers.

^b Normalized Own Weighted Forward Multipliers.

Source: Own elaboration of Tables 2, 3 and 5

key sector that has a predominant role in Iranian Economy. This sector, based on providing foreign exchange required for funding the government budget, has a vital role for Iranian Economy (Salami et al. (2012)). Although the key role of crude oil and natural gas extraction in Iran, based on Power Ministry of Iran (2019), the energy losses is about 182.3 million barrels of oil equivalent in 2011, which is about 38.1% of all energy losses in Iran. This amount of losses is reached to 174.02 million barrels of oil equivalent in 2015, which is equivalent to 34.7% of total energy losses.

These energy losses are due to the natural gas refinery inefficiency and the lack of flared gas collecting system during oil extraction. In addition, unclear reporting of end-use gas produced, and lack of the exact location of its final use due to the limited data and statistics can make the policy recommendation in this sector complicated. While, many suggestions were proposed to reduce the flared gas due to the financial difficulties of the Petroleum Ministry they not have been acted on.

The sale of associated gas to private power plants and the sale of electricity to the national grid, which remains largely unclear due to financial problems between the Petroleum Ministry and the Power Ministry of Iran. Hence, technical and financial policy recommendations can be proposed which significantly reduce the intensity of energy in this sector. Accordingly, we recommended, the creation of surface facilities for the collection of associated natural gases before flaring process, the sale of these gases to small private power plants for free, and the financial settlement of the Petroleum Ministry and the Power Ministry of Iran through the revenue of supplying the electricity to the national grid (Faridzad et al. (2015)).

Regarding the transportation sector, policymakers need to focus on producing efficient energy consumption automobiles especially in the Saipa and Iran Khodro automobile factories, replacing worn-out vehicles in the public transportation, developing rail transport networks such as urban subways, supplying transportation services and managing fuel consumption in private transportation. However, fuel types are also effective from a policy perspective. Moving towards higher-quality fuel production and import of hybrid and electric cars without tariffs is also an appropriate alternative policy (Sarir Afraz et al. (2007) and Nikookar et al. (2017)).

The manufacture of other non-metallic mineral products is a considerably energy-intensive industry in Iran due to the type of technology used. In this sector, commodities such as ceramic and tile based on slurry method is produced. So the energy consumption of non-metallic mineral products in Iran, which uses similar type of equipment and technology production, is very high. Policy solutions such as creating an energy management unit, low cost technical solutions, such as the use of cooling furnaces for heating the production hall, cost-effective solutions, such as removing mercury vapor lamps and replacing them with sodium vapor lamps and cost-effective solutions, such as the replacement of high-efficiency electro motors instead of old ones is recommended (Fazeli et al. (2013)).

The manufacture of chemicals and chemical products in Iran is also one of the most energy intensive industries. More than 15 percent of the energy consumed in this industry is related to cement production. Some technical studies such as Afshar et al. (2015) suggest replacing wet method by dry method and development of pre-heating center as policy recommendations for reducing energy consumption in this sector.

Finally, the services sector including wholesale trade and retail trade, telecommunication services, real-state services, public administration, account for more than 42 percent of Iran's GDP in 2011. According to the Banouei et al. (2009), the services sector in the Iranian Economy is mainly a type of consumption goods sector compared to the other sectors. Consequently, this sector is accompanied with significant consumption-induced effect on

energy intensive commodities. Also, according to Kafaei and Nejadaghaeiavash (2016), it is indicated that the energy intensity of the Iranian services sector is considerable. It seems that trying to change the share of services in Iran and moving towards market-oriented and information technology-based services can reduce energy intensity in this sector.

On the other hand, the type of policies mentioned above as reducing and managing fossil energy intensity directly from an economic sector cannot be recommended for those sectors with high pure weighted backward or forward effects. Regarding the pure effects, more than one sector is involved in fossil energy consumption; So it cannot be determined which economic sector has a greater effect in fossil energy consumption.

To illustrate, Construction activities after services have the highest pure weighted backward effect in energy consumption irrespective of which sector is responsible for energy demand. Hence, policy makers should employ cross-sectoral policies. Similarly, this is the case for high pure weighted forward effects in which more than one sector is responsible for energy consumption in addition to the supply sector i.

Our final finding resembles to the finding of Alcántara et al. (2010) for Spain, which suggests that the Iranian Economy does not have any market-based mechanisms in the form of CO2 taxes or emission trading schemes (ETS) in other energy intensive key sectors. Employing the right prices for energy through economic instruments may not be fully effective to encourage energy intensity in non-energy-intensive sectors. In this study, as recommended by Alcántara et al. (2010), horizontal technologies (applicable across all the production processes within each sector) and process-specific technologies should be recommended for cross-sectoral policy with high pure weighted sectors in Iran.

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